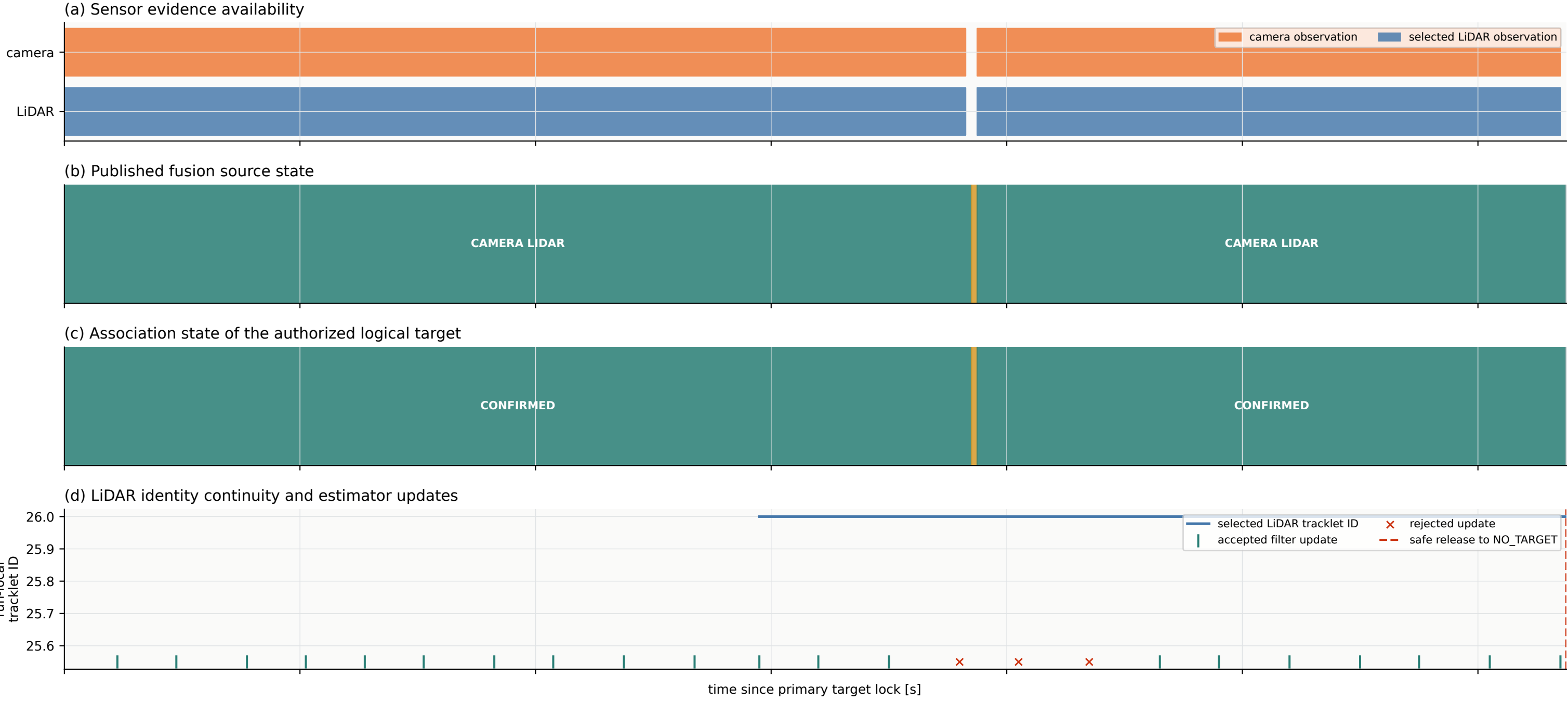




LOGICAL TARGET	EPISODE	SOURCE	INTERPRETATION
run-local ID 1	12.75 s from lock to release	recorded rosbag replay · 0.5×	continuity evidence; no external ground truth